

Solar Irradiance Prediction PoC

Scope, Feature Specification and Sprint Plan (Jira)

Rodic InfraAI Innovation Challenge 2026 · Theme: AI for India Energy Stack

Field	Detail
Version	v2.0 — supersedes v1.0. Scope reset to a simulation-based PoC on public data; no specific plant, state or regulation.
Product owner	Siddarth, Founder & CEO
Engineering	Akshant — Data & ML model · Abhiram — Visual dashboard, hosting & documentation
Sprint	Wednesday 30 Sep → Wednesday 7 Oct 2026 (8 days)
Feature freeze	Monday 5 Oct, 20:00 IST
Submission	Wednesday 7 Oct (external deadline Thursday 8 Oct is buffer only)
Three deliverables	1. ML prediction model · 2. Visual dashboard (hosted, open access) · 3. Video demo, 3 minutes
Jira project	SOLAR · Sprint “PoC-1” · stories SOLAR-6 to SOLAR-32

In one sentence. By 7 October we will submit a machine-learning model that predicts next-day hourly solar irradiance with an honest uncertainty band for five Indian locations, a public dashboard where anyone can check those predictions against what actually happened, and a three-minute video that shows both working.

1. Executive summary

Situation

We have been shortlisted from more than 250 applicants for the Proof-of-Concept round of the Rodic InfraAI Innovation Challenge 2026. Our entry is AI-based solar irradiance prediction. The organisers want a working prototype, a testing report, open documentation and a short video.

Complication

Solar generation swings with cloud cover, so anyone planning around solar power — plant operators, grid planners, industrial users — needs to know tomorrow's sunlight in advance, and how far to trust that estimate. Standard weather forecasts give a single number with no measure of confidence. In a competition, the parallel problem is that most entries will show an accuracy figure that the evaluator cannot verify.

Resolution

We will build a focused, verifiable PoC rather than a broad one:

- **An ML model** that turns the previous evening's weather forecast into next-day hourly irradiance, with a P10–P90 range, and demonstrably beats two simple baselines.
- **A visual dashboard** where the evaluator can pick any location and any day in the test period and see prediction against reality for themselves.
- **A three-minute video** that walks from the problem to the live dashboard to the measured results.

Why this can win

- **Verifiability.** Every accuracy claim can be checked inside the dashboard.
- **Honest uncertainty.** A calibrated range, not just a single number.
- **Rigour an expert will recognise.** The model is trained only on forecasts that existed the evening before, and tested on months it never saw.
- **Deliverable in eight days** with two engineers, using only free, open data that is available today.

2. Scope

Problem statement

Predict next-day hourly global horizontal irradiance (GHI, W/m²) for a given Indian location from public weather data, with a calibrated uncertainty range, and make the predictions and their accuracy visible in a web dashboard.

Prediction task definition

Element	Definition
Forecast issued	Each evening, using only information available at that time
Forecast target	Next day, hourly, 06:00–19:00 IST (14 values per day)
Output	GHI at P10, P50 and P90 (W/m ²); daily irradiation total (kWh/m ² /day) at P10, P50 and P90
Inputs	Day-ahead weather-model forecast (irradiance, cloud cover by layer, temperature, humidity); computed sun position and clear-sky irradiance; recent observed irradiance
Ground truth (“actual”)	Hourly GHI from ERA5 reanalysis, cross-checked against NASA POWER satellite-derived GHI
Locations	Five Indian sites in different climate zones, e.g. Jodhpur, Delhi, Ahmedabad, Nagpur, Chennai (final list confirmed in SOLAR-6)
Training period	January 2024 – December 2025
Test period	January – September 2026, never used for training or tuning

In scope

Deliverable	What it includes	Form question
1. ML prediction model	Data pipeline, two baselines, LightGBM quantile model, calibration, optional confidence flag, evaluation script	Q8, Q9
2. Visual dashboard	Five pages: forecast view, accuracy scorecard, location comparison, energy estimate, data & method. Hosted at a public URL, no login	Q7
3. Video demo (3 min)	Problem, approach, live dashboard walkthrough, results, path to deployment. Unlisted on YouTube	Q11
Supporting documents	Testing report PDF; documentation site with data-privacy note and open-source acknowledgements	Q9, Q10, Q12

Out of scope (not built, not discussed in stand-up)

- Real plant data, a named plant, a specific state or regulator, penalty calculations.
- Live daily forecasting pipeline, scheduled jobs, public REST API, database.
- User accounts or logins; alerts or notifications.
- Deep-learning models (LSTM, transformers). LightGBM is sufficient and explainable.
- Intraday (next-hour) nowcasting.

3. Data plan

Every source below is free, needs no approval or account, and has been confirmed to hold the required history.

Role	Source	Coverage used	Licence / access	Why
Model inputs (day-ahead forecasts)	Open-Meteo Previous Runs API	Jan 2024 → Sep 2026, lead time 1 day	CC BY 4.0; no API key	Fixed day-ahead lead time: the model sees only what was forecast the evening before
Model inputs (backup, longer history)	Open-Meteo Historical Forecast API (GFS)	GFS archive from Mar 2021	CC BY 4.0; no API key	Fallback if Previous Runs lacks a variable
Ground truth	Open-Meteo Historical Weather API (ERA5)	Hourly, gap-free	CC BY 4.0; no API key	Consistent hourly GHI for every site
Cross-check of ground truth	NASA POWER hourly API	Hourly GHI	Public; no API key	Independent satellite-based check on ERA5
Sun position & clear-sky GHI	pvlib (computed)	Any time, any place	BSD-3 licence	Converts GHI to a clear-sky index, which is easier to model

Known limitation (to be stated openly in the testing report). ERA5 and NASA POWER are modelled or satellite-derived, not ground-station measurements. This is acceptable for a PoC; a production deployment would train on on-site pyranometer data.

Output file contract between Akshant and Abhiram (frozen Wed 30 Sep)

File	Contents
sites.json	site_id, name, latitude, longitude, climate zone
forecasts.parquet	One row per site per hour: site_id, date, hour_ist, ghi_actual, ghi_p10, ghi_p50, ghi_p90, ghi_nwp (raw weather model), ghi_persistence, ghi_clearsky, confidence (High/Medium/Low or null), split (train/test)
metrics.json	For each site x model x month: MAE, RMSE, nRMSE, bias, skill vs persistence, skill vs raw NWP, P10-P90 coverage; plus overall test-period totals
energy.py	Function daily_energy(site_id, date, capacity_kw) → P10 / P50 / P90 kWh

Abhiram builds against a sample version of these files from Day 1; Akshant replaces them with real outputs on Sun 4 Oct. Any change to the contract needs both engineers to agree.

4. Feature specification

Priority rule. P0 = the submission is not credible without it. P1 = strengthens the entry; cut if behind schedule. **Cut order if the Sunday 4 Oct checkpoint is missed:** F3, then F7, then F6.

ID	Feature	Deliverable	Priority	Owner	Due
F1	Day-ahead probabilistic forecast model	ML model	P0	Akshant	Sun 4 Oct
F2	Baseline benchmarking	ML model	P0	Akshant	Fri 2 Oct
F3	Forecast confidence flag	ML model	P1	Akshant	Sun 4 Oct
F4	Forecast view (prediction vs actual)	Dashboard	P0	Abhiram	Sun 4 Oct
F5	Accuracy scorecard	Dashboard	P0	Abhiram	Sun 4 Oct
F6	Location comparison	Dashboard	P1	Abhiram	Mon 5 Oct
F7	Energy estimate calculator	Dashboard	P1	Akshant (logic), Abhiram (UI)	Mon 5 Oct
F8	Data & method page	Dashboard	P0	Abhiram	Mon 5 Oct
F9	Three-minute video demo	Video	P0	Siddarth (with both)	Tue 6 Oct

Deliverable 1 — ML prediction model

F1 · Day-ahead probabilistic forecast model

P0 · ML model

Owner: Akshant · **Due:** Sun 4 Oct

Why it matters. This is the core of the entry. A range (P10–P90) is more useful to an operator than a single number, and it is what distinguishes us from a plain weather forecast.

User story. As a solar operator, I want tomorrow's hourly irradiance with a likely range, so that I can plan for both a good and a poor day.

Specification

- Target: clear-sky index (actual GHI ÷ clear-sky GHI), converted back to GHI after prediction. This removes the daily sun cycle so the model learns only the effect of weather.
- Algorithm: LightGBM quantile regression, three models at quantiles 0.1, 0.5 and 0.9, trained on all five sites together with site as a feature.
- Features: forecast GHI, low/mid/high cloud cover, temperature, relative humidity, sun elevation, clear-sky GHI, day-of-year (sine and cosine), previous day's observed clear-sky index.
- Calibration: adjust the band on 2025 data so that about 80% of actual values fall inside P10–P90.
- Validation: time-based only. Train 2024, tune on 2025, retrain on 2024–2025, report on 2026.

Acceptance criteria

- On the 2026 test period, P50 has lower MAE than both baselines (F2) across all five sites combined.
- P10–P90 coverage on the test period is between 75% and 85%.
- $P10 \leq P50 \leq P90$ in every row; night-time values are zero.
- No random train/test split anywhere in the code; reviewed by Abhiram.

F2 · Baseline benchmarking**P0** · ML model**Owner:** Akshant · **Due:** Fri 2 Oct

Why it matters. An accuracy number means nothing without a comparison. Beating credible baselines is the evidence an expert evaluator looks for first.

User story. As an evaluator, I want to see whether the ML model adds value over simple methods.

Specification

- Baseline A — persistence: tomorrow's clear-sky index equals today's.
- Baseline B — raw weather model: the weather model's own GHI forecast, unmodified.
- Both are scored with exactly the same metric code as F1.

Acceptance criteria

- Both baselines are present in forecasts.parquet and metrics.json for every site and test month.

F3 · Forecast confidence flag**P1** · ML model**Owner:** Akshant · **Due:** Sun 4 Oct

Why it matters. Tells the user which days to trust. It also carries our Evidence Contradiction Graph idea — disagreement between sources signals risk — into the energy domain.

User story. As an operator, I want to know which days the forecast is less reliable, so I can plan more cautiously on those days.

Specification

- Measure how much two independent weather models disagree on tomorrow's irradiance, plus the width of the P10–P90 band.
- Map to High, Medium or Low confidence using thresholds set on 2025 data.

Acceptance criteria

- On the 2026 test period, average error on Low-confidence days is clearly higher than on High-confidence days.
- If this test fails, the feature is dropped from the dashboard and noted in the testing report. It is never shown if it does not work.

Deliverable 2 — Visual dashboard

Built in Streamlit, hosted free on Streamlit Community Cloud (or Hugging Face Spaces), public URL, no login. The dashboard reads the output files only; it has no backend, database or live data feed. It must load in under 5 seconds and work on a phone.

F4 · Forecast view: prediction vs actual

P0 · Dashboard

Owner: Abhiram · **Due:** Sun 4 Oct

Why it matters. This is the page the evaluator will remember. It lets them test the model on any day they choose, including difficult cloudy days.

User story. As an evaluator, I pick a location and a date and see what the model predicted the evening before, next to what actually happened.

Specification

- Controls: site selector, date picker limited to the test period, and a shortcut button “Show a cloudy day”.
- Chart: hourly P50 line, shaded P10–P90 band, actual GHI as dots, raw weather-model forecast as a dashed line.
- Cards above the chart: predicted daily irradiation (P10 / P50 / P90), actual daily irradiation, error %, confidence flag (if F3 ships).
- Labelled clearly: “Test period — the model never saw this data during training.”

Acceptance criteria

- Every site and every test date renders without error.
- Values on the chart match forecasts.parquet exactly.
- Page is readable on a mobile screen.

F5 · Accuracy scorecard

P0 · Dashboard

Owner: Abhiram · **Due:** Sun 4 Oct

Why it matters. Turns the testing report into one visual page an evaluator can absorb in 30 seconds.

User story. As an evaluator, I want one page that shows how accurate the model is and how it compares with simpler methods.

Specification

- Headline sentence generated from metrics.json, e.g. “Across five sites in 2026, the model's error was X% lower than the raw weather forecast.”
- Bar chart: MAE and RMSE for the ML model, persistence and raw weather model.
- Monthly error line chart, showing monsoon months honestly.
- Calibration chart: expected vs actual coverage of the P10–P90 band.

Acceptance criteria

- No number on the page is typed by hand; all are read from metrics.json.
- Numbers match the testing report exactly.

F6 · Location comparison

P1 · Dashboard

Owner: Abhiram · **Due:** Mon 5 Oct**Why it matters.** Shows the model generalises across India's climate zones rather than working in one place only.**User story.** As an evaluator, I want to see whether the model works equally well across different climates.**Specification**

- Map of the five sites, each coloured by test-period error.
- Table: per-site MAE, skill vs raw weather model, coverage.
- Heatmap: error by site × month.

Acceptance criteria

- All five sites shown; clicking a site opens F4 for that site.

F7 · Energy estimate calculator

P1 · Dashboard

Owner: Akshant (logic), Abhiram (UI) · **Due:** Mon 5 Oct**Why it matters.** Translates irradiance, which is abstract, into energy, which an industrial user understands.**User story.** As a plant planner, I enter a system size and see the likely energy output for a selected day.**Specification**

- Inputs: site, date, system capacity (kW or MW).
- Output: expected daily energy at P10 / P50 / P90 and an hourly power chart.
- Uses pvlib with standard assumptions (fixed tilt equal to latitude, south-facing, typical losses), all shown on the page.

Acceptance criteria

- Doubling the capacity doubles the output.
- Assumptions are visible on the page and cannot be hidden.

F8 · Data & method page

P0 · Dashboard

Owner: Abhiram · **Due:** Mon 5 Oct**Why it matters.** Answers the evaluator's questions about data sources, licences, privacy and approach before they ask.**User story.** As an evaluator, I want to know what data was used, how the model works and what its limits are.**Specification**

- Data sources with licence and attribution; the training and test windows.
- A one-diagram explanation of the method: weather forecast → features → quantile model → calibrated range.
- Privacy statement: only public weather data; no personal data collected or processed.
- Limitations and a short “path to production” paragraph: on-site sensor data, daily automated forecasts, API integration.

Acceptance criteria

- Every source carries its attribution text as required by its licence.

Deliverable 3 — Video demo

F9 · Three-minute video demo

P0 · Video

Owner: Siddarth (presenter), Abhiram (screen recording), Akshant (results) · **Due:** Tue 6 Oct

Why it matters. Many evaluators will watch the video before they open the dashboard. It must stand on its own.

User story. As an evaluator, I want to understand the problem, the solution and the evidence in three minutes.

Specification

- Length: 2:45–3:00. Unlisted on YouTube (viewable by link, not private).
- Structure and script outline: section 7.
- Screen recording of the live hosted dashboard, not a local copy.
- Every number spoken matches metrics.json.

Acceptance criteria

- Runs under 3:00.
- Opens from the link in an incognito browser.
- Clear audio; readable screen text at 1080p.

5. Sprint set-up and epics (Jira)

Field	Value
Sprint	SOLAR PoC-1 · Wed 30 Sep → Wed 7 Oct 2026
Sprint goal	Submit a verified ML irradiance model, a public dashboard and a 3-minute video by Wednesday 7 October.
Ceremonies	Stand-up 09:30 (10 min) · Evening demo 19:30 (15 min, show what works on the hosted URL) · Checkpoint Sun 4 Oct 19:30 · Feature freeze Mon 5 Oct 20:00
Estimation	1 story point $\approx$ 1 focused hour
Board	To Do → In Progress → In Review → Done

Epics

Epic	Name	Outcome	Lead	Stories	Points
SOLAR-1	Data Foundation	Clean, aligned, licensed data for five sites	Akshant	SOLAR-6 – SOLAR-10	13
SOLAR-2	ML Prediction Model	Calibrated quantile model that beats both baselines	Akshant	SOLAR-11 – SOLAR-16	17
SOLAR-3	Visual Dashboard	Five-page hosted dashboard on real model outputs	Abhiram	SOLAR-17 – SOLAR-24	22
SOLAR-4	Evaluation & Documentation	Testing report, dataset description, documentation site	Both	SOLAR-25 – SOLAR-28	11
SOLAR-5	Video Demo & Submission	3-minute video; verified links; form submitted	Siddarth	SOLAR-29 – SOLAR-32	8

Capacity check

Person	Points	Working days	Average per day	Verdict
Akshant	37	7	$\approx$ 5.3 h	Comfortable; leaves time for model iteration
Abhiram	28	7	$\approx$ 4.0 h	Comfortable; buffer for hosting issues
Siddarth	6	—	—	Script, video, submission, reviews

Shared stories are counted against the first-named owner.

Definition of Done

- Merged to main via a pull request reviewed by the other engineer.
- Shown working on the **hosted URL** at the evening demo, not on a laptop.
- Any number displayed or spoken comes from the evaluation script.
- Any open-source library used is added to the acknowledgements list.

6. Jira stories

Epic SOLAR-1 · Data Foundation

SOLAR-6 · Spike: confirm sites, variables and data availability

Story · F1

Epic SOLAR-1 **Assignee** Akshant **Points** 2 **Priority** Highest **Due** Wed 30 Sep **Blocks** SOLAR-7, 8

Description. Before writing the pipeline, confirm the five site coordinates and that every variable exists for Jan 2024 – Sep 2026 at a 1-day lead time.

Acceptance criteria

- A one-page note lists each variable, its source and its first available date, shared by 19:30.
- If any variable is missing from the Previous Runs API, the note names the fallback source.

Sub-tasks

- Test-pull one week for one site from each API.
- Finalise five site coordinates.

SOLAR-7 · Ingest day-ahead weather forecasts

Story · F1

Epic SOLAR-1 **Assignee** Akshant **Points** 3 **Priority** Highest **Due** Thu 1 Oct **Blocks** SOLAR-9

Description. Download day-ahead forecasts for all sites and variables, Jan 2024 – Sep 2026.

Acceptance criteria

- **Given** any site and date, **then** the stored value is the forecast made the day before, not a later update.
- Re-running the download does not create duplicates.

Sub-tasks

- Download script with retries.
- Save raw data per site as Parquet.

SOLAR-8 · Ingest actual irradiance and cross-check

Story · F1

Epic SOLAR-1 **Assignee** Akshant **Points** 3 **Priority** Highest **Due** Thu 1 Oct **Blocks** SOLAR-9

Description. Download hourly ERA5 GHI as ground truth and NASA POWER GHI as an independent check.

Acceptance criteria

- Daily totals from the two sources agree closely on clear days; any site with large disagreement is flagged in the note.

SOLAR-9 · Clean data, align time zones, quality checks

Story · F1

Epic SOLAR-1 **Assignee** Akshant **Points** 3 **Priority** Highest **Due** Fri 2 Oct **Blocks** SOLAR-10

Description. Convert all timestamps to IST, remove impossible values, zero night hours.

Acceptance criteria

- On a known clear day, actual and clear-sky curves peak in the same hour.
- A short report counts rows removed and why.

SOLAR-10 · Build feature tableStory · F1

Epic SOLAR-1	Assignee Akshant	Points 2	Priority High	Due Fri 2 Oct	Blocks SOLAR-11, 12
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Description. Compute sun position, clear-sky GHI and clear-sky index with pvlib; assemble all model features.

Acceptance criteria

- One table covering all sites, with train/tune/test split column.

Epic SOLAR-2 · ML Prediction Model

SOLAR-11 · Baseline modelsStory · F2

Epic SOLAR-2	Assignee Akshant	Points 2	Priority Highest	Due Fri 2 Oct	Blocks SOLAR-25
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Description. Persistence and raw weather-model baselines.

Acceptance criteria

- Both baselines scored on 2025 and 2026 with the shared metric code.

SOLAR-12 · Train LightGBM quantile modelStory · F1

Epic SOLAR-2	Assignee Akshant	Points 5	Priority Highest	Due Sat 3 Oct	Blocks SOLAR-13, 15
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Description. Three quantile models (0.1, 0.5, 0.9) on clear-sky index; tune on 2025; retrain on 2024-2025.

Acceptance criteria

- P50 beats both baselines on the 2025 tuning set; result reported at the Saturday evening demo.

Sub-tasks

- Hyper-parameter search on 2025 only.
- Save models with version tag.

SOLAR-13 · Calibrate the P10-P90 bandStory · F1

Epic SOLAR-2	Assignee Akshant	Points 3	Priority High	Due Sun 4 Oct	Blocks SOLAR-15
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Description. Adjust the range so it contains about 80% of actual outcomes, and prevent P10 > P50 or P50 > P90.

Acceptance criteria

- Coverage on 2026 test data between 75% and 85%.
- Zero quantile crossings.

SOLAR-14 · Confidence flag (P1)Story · F3

Epic SOLAR-2	Assignee Akshant	Points 3	Priority Medium	Due Sun 4 Oct	Blocks —
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Description. Score disagreement between two weather models; set High/Medium/Low thresholds on 2025.

Acceptance criteria

- Low-confidence days show clearly higher error on 2026; if not, the flag is dropped and recorded in the decision log.

SOLAR-15 · Export final model outputs

Story · F1

Epic SOLAR-2 **Assignee** Akshant **Points** 2 **Priority** Highest **Due** Sun 4 Oct **Blocks** SOLAR-19, 20**Description.** Write sites.json, forecasts.parquet and metrics.json to the agreed contract; hand over to Abhiram by 14:00.**Acceptance criteria**

- Files validate against the contract; dashboard loads them without code changes.

SOLAR-16 · Energy estimate function

Story · F7

Epic SOLAR-2 **Assignee** Akshant **Points** 2 **Priority** Medium **Due** Mon 5 Oct **Blocks** SOLAR-22**Description.** daily_energy(site_id, date, capacity_kw) using pvlib with stated standard assumptions.**Acceptance criteria**

- Output scales linearly with capacity; returns P10 / P50 / P90 kWh and hourly kW.

Epic SOLAR-3 · Visual Dashboard**SOLAR-17 · Repo, output contract, Streamlit skeleton**

Story · F4

Epic SOLAR-3 **Assignee** Abhiram (+ Akshant for contract) **Points** 3 **Priority** Highest **Due** Wed 30 Sep **Blocks** SOLAR-18**Description.** Public GitHub repo; freeze the file contract with Akshant; skeleton app with five empty pages on sample files.**Acceptance criteria**

- Contract committed with a sample file set.
- App runs locally with all five page tabs.

SOLAR-18 · Deploy to a public URL

Story · F4

Epic SOLAR-3 **Assignee** Abhiram **Points** 2 **Priority** Highest **Due** Thu 1 Oct **Blocks** All UI stories**Description.** Deploy the skeleton now so hosting problems appear early.**Acceptance criteria**

- URL opens on a phone and in incognito without login.

SOLAR-19 · Forecast view page

Story · F4

Epic SOLAR-3 **Assignee** Abhiram **Points** 5 **Priority** Highest **Due** Sun 4 Oct **Blocks** SOLAR-30**Description.** Build on sample data from Fri 2 Oct; switch to real outputs on Sun 4 Oct.**Acceptance criteria**

- All F4 acceptance criteria pass on real data.

Sub-tasks

- Site and date controls.
- Chart with band, actuals, raw forecast.
- Summary cards.
- “Show a cloudy day” button.

SOLAR-20 · Accuracy scorecard page						Story · F5
Epic SOLAR-3	Assignee Abhiram	Points 3	Priority Highest	Due Sun 4 Oct	Blocks SOLAR-30	
Description. Headline, comparison bars, monthly error, calibration chart.						
Acceptance criteria						
All F5 acceptance criteria pass.						
SOLAR-21 · Location comparison page (P1)						Story · F6
Epic SOLAR-3	Assignee Abhiram	Points 3	Priority Medium	Due Mon 5 Oct	Blocks —	
Description. Map, per-site table, site x month heatmap.						
Acceptance criteria						
All F6 acceptance criteria pass.						
SOLAR-22 · Energy estimate page (P1)						Story · F7
Epic SOLAR-3	Assignee Abhiram	Points 2	Priority Medium	Due Mon 5 Oct	Blocks —	
Description. Capacity input, P10/P50/P90 energy, hourly chart, assumptions box.						
Acceptance criteria						
All F7 acceptance criteria pass.						
SOLAR-23 · Data & method page						Story · F8
Epic SOLAR-3	Assignee Abhiram	Points 2	Priority High	Due Mon 5 Oct	Blocks SOLAR-28	
Description. Sources and licences, method diagram, privacy statement, limitations, path to production.						
Acceptance criteria						
All F8 acceptance criteria pass.						
SOLAR-24 · Visual polish, mobile and speed						Story · F4-F8
Epic SOLAR-3	Assignee Abhiram	Points 2	Priority High	Due Mon 5 Oct	Blocks SOLAR-30	
Description. Consistent colours, readable fonts, Sahasranshu branding, caching for speed.						
Acceptance criteria						
- Every page loads in under 5 seconds on the hosted URL. - No page breaks on a phone.						
Epic SOLAR-4 · Evaluation & Documentation						
SOLAR-25 · Evaluation script (single source of truth)						Story · F1, F2, F5
Epic SOLAR-4	Assignee Akshant	Points 3	Priority Highest	Due Mon 5 Oct	Blocks SOLAR-26	
Description. One script produces every metric used in the dashboard, report and video.						
Acceptance criteria						
- Writes metrics.json and the tables used in the testing report. - Re-running gives identical numbers.						

SOLAR-26 · Testing report PDF (Q9)

Story · F1, F2

Epic SOLAR-4 **Assignee** Akshant **Points** 3 **Priority** Highest **Due** Tue 6 Oct **Blocks** SOLAR-32**Description.** Accuracy vs baselines, calibration, by site and month, confidence-flag result, limitations.**Acceptance criteria**

- Open-access link works in incognito.
- Every number matches metrics.json.

SOLAR-27 · Dataset description text (Q8)

Story · —

Epic SOLAR-4 **Assignee** Akshant **Points** 1 **Priority** High **Due** Tue 6 Oct **Blocks** SOLAR-32**Description.** Short answer covering source, size (rows, date range, sites) and key features.**Acceptance criteria**

- Under 200 words; reviewed by Siddarth.

SOLAR-28 · Documentation site (Q10, Q12)

Story · F8

Epic SOLAR-4 **Assignee** Abhiram **Points** 4 **Priority** Highest **Due** Tue 6 Oct **Blocks** SOLAR-32**Description.** GitHub README or GitHub Pages: approach, architecture, tools and APIs, data privacy, open-source acknowledgements with licences, how to run.**Acceptance criteria**

- Open-access link works in incognito.
- Every library used is listed with its licence.

Epic SOLAR-5 · Video Demo & Submission**SOLAR-29 · Video script and storyboard**

Story · F9

Epic SOLAR-5 **Assignee** Siddarth **Points** 2 **Priority** High **Due** Mon 5 Oct **Blocks** SOLAR-30**Description.** Word-for-word script of about 400 words, timed to section 7.**Acceptance criteria**

- Read-aloud time under 2:50.

SOLAR-30 · Record and edit 3-minute video

Story · F9

Epic SOLAR-5 **Assignee** Siddarth + Abhiram **Points** 3 **Priority** Highest **Due** Tue 6 Oct **Blocks** SOLAR-32**Description.** Screen-record the hosted dashboard; voice-over; upload unlisted to YouTube.**Acceptance criteria**

- All F9 acceptance criteria pass.

SOLAR-31 · Final QA of all links and demo

Story · All

Epic SOLAR-5 **Assignee** Abhiram **Points** 2 **Priority** Highest **Due** Wed 7 Oct **Blocks** SOLAR-32**Description.** Test the dashboard, report, documentation and video links in incognito and on a phone.**Acceptance criteria**

- All five links open without login.

SOLAR-32 · Submit the PoC form

Story · All

Epic SOLAR-5

Assignee Siddarth

Points 1

Priority Highest

Due Wed 7 Oct

Blocks —

Description. Complete Q7-Q12 and submit; save the printed copy.

Acceptance criteria

- Submission confirmation saved by 18:00 Wed 7 Oct.

7. Video demo plan (3 minutes)

Target 2:50, hard limit 3:00. One presenter voice-over on a screen recording of the hosted dashboard. Script about 400 words, speaking at a steady pace.

Time	Section	On screen	Message
0:00-0:20	Hook	Title card; a day's irradiance curve dropping as cloud arrives	Solar output can halve in an hour when clouds move in. Anyone relying on solar power needs to know tomorrow's sunlight — and how sure that estimate is.
0:20-0:40	Problem	Weather forecast single line vs actual	Standard weather forecasts give one number and no confidence. Operators over- or under-plan.
0:40-1:05	Approach	Method diagram from F8	Public weather forecasts → ML model trained on two years across five Indian climate zones → next-day hourly prediction with a calibrated range.
1:05-2:05	Live demo	Dashboard: F4 clear day, then “Show a cloudy day”; F6 map; F7 capacity input	Pick any site and day — including data the model never saw. The band widens on uncertain days. The same forecast becomes an energy estimate for any system size.
2:05-2:30	Results	F5 scorecard	Measured on nine months of unseen 2026 data: error X% below the raw weather forecast; the range captures Y% of outcomes. Say where it is weaker (monsoon).
2:30-2:50	Deployment path	Architecture slide	Ready to plug in on-site sensor data, run automatically each evening and serve forecasts through an open API — aligned with the India Energy Stack.
2:50-3:00	Close	Logo, dashboard URL	Sahasranshu Technologies. Thank you.

Production notes. Record at 1080p, browser zoom 110%, hide bookmarks bar, close notifications. Replace X and Y with values from metrics.json on Tue 6 Oct. Upload as **unlisted**, not private, and test the link in incognito.

8. Day-by-day plan

Day	Akshant	Abhiram	Siddarth	End-of-day check (19:30)
Wed 30 Sep	SOLAR-6 data spike	SOLAR-17 repo, contract (joint), skeleton	Approve sites and scope	Contract frozen; data confirmed
Thu 1 Oct	SOLAR-7, 8 data downloads	SOLAR-18 deploy public URL; start SOLAR-19 on sample	—	Public URL live; raw data for all sites
Fri 2 Oct	SOLAR-9, 10 clean + features; SOLAR-11 baselines	SOLAR-19 forecast view on sample data	—	Baseline scores shared
Sat 3 Oct	SOLAR-12 quantile model	SOLAR-20 scorecard on sample; SOLAR-23 draft	—	Model beats baselines on 2025
Sun 4 Oct	SOLAR-13 calibration; SOLAR-14 flag; SOLAR-15 outputs by 14:00	Switch SOLAR-19, 20 to real outputs	Checkpoint review	CHECKPOINT: F4 and F5 working on real data; else apply cut order
Mon 5 Oct	SOLAR-16 energy; SOLAR-25 evaluation	SOLAR-21, 22, 23, 24	SOLAR-29 script	Feature freeze 20:00
Tue 6 Oct	SOLAR-26 report; SOLAR-27 Q8 text	SOLAR-28 documentation; SOLAR-30 recording	SOLAR-30 video	Report, docs and video links ready
Wed 7 Oct	Review and fixes	SOLAR-31 final QA	SOLAR-32 submit	Submitted; confirmation saved

Checkpoint rule, Sunday 4 October, 19:30. If a stranger cannot open the public URL and compare the model's prediction with the actual values for any site and test date, we apply the cut order (F3, then F7, then F6) that evening and put all effort into F4, F5 and the video.

9. Risks and decisions

Risk	Likelihood	Impact	Mitigation	Owner
A needed variable is missing at 1-day lead time	Medium	Medium	SOLAR-6 identifies it on Day 1; fall back to the Historical Forecast API (GFS)	Akshant
Model does not beat the raw weather forecast everywhere	Medium	High	Report honestly by site and month; lead with calibration and verifiability	Akshant
Monsoon months show high error	Certain	Low	Show openly in F5; explain the physics; the range widens on those days	Akshant
Hosting or dependency problems late in the week	Low	High	Deploy on Thu 1 Oct; pin library versions	Abhiram
Video overruns 3 minutes	Medium	Medium	Script timed on Mon 5 Oct; cut the location map segment first	Siddarth
Scope creep	Medium	High	Out-of-scope list in section 2; new ideas go to a “later” list	Siddarth

Decision log

#	Decision	Status
D1	Simulation-based PoC on public data; no named plant, state or regulation	Decided
D2	Three deliverables: ML model, visual dashboard, 3-minute video	Decided
D3	LightGBM quantile regression; no deep learning	Decided
D4	Streamlit dashboard hosted on Streamlit Community Cloud (fallback: Hugging Face Spaces)	Decided
D5	Final five sites	Confirm Wed 30 Sep (SOLAR-6)
D6	Confidence flag ships or is dropped	Decided by SOLAR-14 test, Sun 4 Oct

Note to the team. This is deliberately narrow. A focused model that honestly beats its baselines, a dashboard that lets the evaluator check it, and a crisp three-minute story will beat a broad entry that cannot be verified. Protect the Sunday checkpoint and the Monday freeze.